

RFI — Stall audio capture for equine monitoring (BSV EquiCare)

To: Industrial / network / bioacoustic audio vendor

From: Bharat Sports Venture — EquiCare engineering

Covers client monitoring point: 8 (stable vices — weaving, crib-biting, wind-sucking), plus respiratory sound (cough) supporting points 3–4

0. How to reply — please read first

Please answer all of this in ONE consolidated reply. We are evaluating several suppliers in parallel on a fixed timeline and will not be sending rounds of follow-up questions. In practice that means:

1. **Answer every numbered question**, in the reply slots provided. If something is not supported, **write that plainly** — "not supported" is a perfectly good answer and costs you nothing. A blank or skipped item, however, we have to record as *not supported*, because we cannot tell the difference.
2. **Attach the documents and files listed in the checklist at the end.** Please send them with this reply, not "on request" — a claim we cannot verify counts as unverified.
3. **Send the actual artifact, not a description of it.** Our single worst experience with a previous supplier: they confirmed "Linux SDK — yes", and the package that arrived was x86-64 only, with no ARM64 build available at all. That cost both sides weeks. So where we ask for a binary, driver, register map or protocol spec, please attach the real file (or a download link) — or state clearly that it does not exist.
4. If a question does not apply to your product, write **"N/A"** and one line saying why.

We would rather receive an honest reply with several "not supported" answers than an optimistic one that unravels at integration.

1. What we are building

A 24/7 monitoring system for stabled horses in India. Each stall has sensors feeding an **on-site edge computer** (NVIDIA Jetson). We need **per-stall audio** so our models can detect specific acoustic events:

- **crib-biting** — teeth on wood/metal: a sharp, broadband transient
- **wind-sucking** — a characteristic grunt/gulp
- **coughing** — respiratory irritation
- **weaving** — rhythmic footfall and rocking

All detection runs on our edge box. We need clean, uncompressed, timestamped audio — not your sound-classification output.

2. Hard requirements — please confirm these first

Pass/fail for us. If any cannot be met, please say so plainly rather than leaving it blank.

#	Requirement	Why
H1	Uncompressed PCM (or lossless) audio available to us — not only AAC / Opus / G.711.	This is the single most important item. Lossy voice codecs destroy exactly the transients — teeth-on-wood, cough onset — that our models depend on.
H2	AGC, noise suppression, echo cancellation and any noise gate can be FULLY DISABLED, in favour of fixed/manual gain.	AGC flattens the level information we use to separate a cough from distant noise; a noise gate would delete quiet breathing entirely. Voice-optimised DSP actively harms us.
H3	Linux ARM64 (aarch64 / NVIDIA Jetson, Ubuntu 22.04+). Either a standard ALSA/PipeWire device or a documented C/C++/Python API.	Our edge box is a Jetson. A previous vendor shipped x86-only binaries with no Python binding and could not rebuild — weeks lost.
H4	No mandatory vendor cloud; no raw audio off-site.	Stall audio can capture staff speech. Under India DPDP-2023 it must stay on-site with configurable retention and no undisableable phone-home.
H5	Any on-device analytics can be disabled — the raw stream must always be available.	We run inference ourselves.

REPLY (H1-H5)

3. Acoustic performance

Please answer with **specific values / part numbers**, not "yes — supported".

1. **Transducer type** — analog or PDM MEMS, electret, condenser — and which you recommend for an unattended Indian stall (0–50 °C, high humidity, dust, ammonia). **Part number + datasheet.**

REPLY

2. **Full frequency response curve** with tolerance. We need usable response across **50–500 Hz** (respiration, grunt, crib wood-contact), **500 Hz–4 kHz** (cough, wind-suck gulp) and up to **8–12 kHz** (teeth-on-wood

transients).

REPLY

3. **Sensitivity and maximum SPL / Acoustic Overload Point.** The same channel must capture a quiet ~30 dB breath at 2–3 m *and* a >100 dB crib-bite or whinny **without clipping**. Please state both ends.

REPLY

4. **Self-noise (EIN, dB-A) and SNR** — for the realistic **mic + preamp + ADC** chain, not the capsule alone. A resting horse breathing at 2–3 m is a very low-level signal.

REPLY

5. **Effective pickup radius** — will it reliably catch quiet breathing and crib-biting **anywhere in a ~3.5 × 3.5 m stall**? State the recommended mounting height and distance.

REPLY

6. **Cross-talk between adjacent stalls.** State the polar pattern (omni / cardioid / shotgun / beamforming array) and **off-axis rejection figures in dB** — a cough from the next stall must not be misattributed to this horse.

REPLY

7. **Sound-source localisation / direction-of-arrival**, if you offer a multi-mic array — angular resolution and capsule count.

REPLY

8. **DSP in the signal path** — high-pass, compression, limiting, noise gate. Can **each** be disabled individually? Please be specific; a limiter left on will distort our peaks.

REPLY

9. **Wind, vibration and structure-borne noise mitigation** — windscreen, shock mount, high-pass. Stall walls transmit impact noise, so this matters.

REPLY

4. Digital audio and interfacing

10. **Sample rates and bit depths** available and configurable. We require **≥16 kHz** (32 or 48 kHz preferred) at **16-bit or better**.

REPLY

11. **How does PCM reach the edge box?** State the transport — analog line, **USB Audio (UAC)**, **I2S/PDM**, **PoE network mic (RTSP/RTP)**, ONVIF audio, Dante/AES67 — with cable type and maximum run for each. **Please attach the interface documentation**. If RTSP, confirm an **uncompressed/lossless (L16/PCM)** payload option.

REPLY

12. **Channel count per device**, and whether **multiple mics can be captured synchronised on a common clock**. Is an aggregation box needed, or does each terminate at the Jetson?

REPLY

13. **Per-packet metadata** — timestamp, stall/channel ID, gain setting, **clip flag**, sample rate. We want to carry these into our data contract.

REPLY

14. **End-to-end latency and jitter**, and the buffering/packet size.

REPLY

5. Time synchronisation

15. **Per-sample or per-packet timestamping** — is it done at the device? State **NTP and/or PTP** support with accuracy and jitter. *A detected cough must line up with the video frame that shows which horse coughed, and with the thermal respiration trace — we are targeting tens of milliseconds.*

REPLY

16. **Local store-and-forward** if the link drops — how long, and is it backfilled **without timestamp gaps**?

REPLY

6. Environment, power and mounting

17. **IP65+ and IK (impact) rating**. Confirm sealing against **ammonia/urine vapour**, dust and wash-down, and **chew/tamper resistance**. Critically: **how is the acoustic port protected without destroying high-frequency response?** Please explain the approach.

REPLY

18. **Operating temperature and humidity; 24/7 rating; MTBF.** Include **anti-condensation/anti-fungal** provision for monsoon conditions.

REPLY

19. **Power** — PoE (state class and watts), USB or DC, and draw per unit. *Note it shares the barn PoE + UPS budget with our cameras.*

REPLY

20. **Mounting** for stall ceiling or wall at **2.5–3.5 m**, out of the horse's reach. Brackets available? Impact-resistant housing?

REPLY

7. Security and privacy

21. **Encryption in transit** (TLS / SRTP) and at rest; authentication; per-stream isolation if you host anything. Signed OTA firmware?

REPLY

22. **DPDP-2023 specifics** — confirm audio can be processed and retained **entirely on-site**, that retention is configurable with auto-delete, that access is logged, and that **nothing requires a non-India cloud** (including OTA and telemetry). If you support **on-device feature-only or redacted output**, please describe it.

REPLY

8. Evaluation

We would like **2 evaluation microphones plus any required interface or PoE injector**, with the **driver or interface documentation and sample code** for Linux/ARM64, so we can validate real recording quality on our Jetson bench.

REPLY (AVAILABILITY + LEAD TIME FOR EVALUATION HARDWARE)

9. Optional — references and recommendation

If you have deployed audio in **livestock, veterinary, bioacoustic or noisy-industrial** settings, we would welcome your recommendation on mic model and placement for a 3.5 × 3.5 m stall, plus any **sample recordings** from a comparable environment.

REPLY

Response checklist

Please confirm each item is attached or answered, so we can evaluate in a single pass:

	Item
☐	Every numbered question answered (or marked "not supported" / "N/A")
☐	All hard requirements (H-numbers) explicitly confirmed or declined
☐	Sensor / component datasheets
☐	Protocol document, register map or API reference — the actual file
☐	ARM64 (aarch64) build attached or download link — or a plain statement that none exists

	Item
☐	Sample data payload showing real field names and units
☐	Sample code, if you provide any
☐	Evaluation-unit availability and lead time
☐	A named technical contact we can reach directly with any single clarification

If anything on this list genuinely cannot be shared before an order, please say which item and why, so we can factor it in rather than assume the worst.

Commercial terms are not part of this RFI — we will discuss those separately once the technical fit is confirmed.